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# OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

## LECTURE 25: MODELLING ARRIVAL AND SERVICE PROCESSES

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- The input or arrival process.** Arrivals are called *customers*. Sometimes *bulk arrivals* rather than *individual arrivals* are applicable. Models in which arrivals are drawn from a small population are called *finite source models*. Usually a queue can accommodate only a finite number of customers. What happens in this case to customers who arrive when the queue is full? We may have the case of *customers who return later* or of *balked customers* (lost customers).
- The output or service process.** A *service time distribution* governs the duration of each customer's service (this may be deterministic in some cases). It must also be specified whether the servers are organised *in parallel* or *in series*.

## The queue discipline.

FCFS = first come, first served,

LCFS = last come, first served,

SIRO = service in random order,

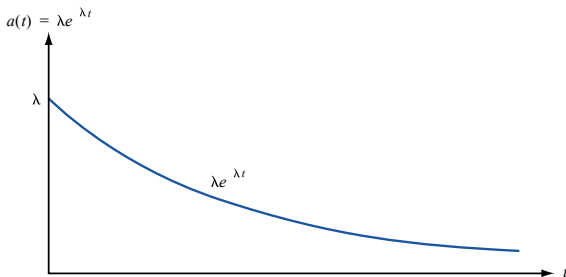
Priority queuing disciplines may also be employed.

The method used by arrivals to join the queue. Sometimes customers join a single queue, whereas sometimes there are several queues and customers then join the shortest queue. Sometimes it is hard to determine which queue is shortest — and can customers then switch, or jockey, between queues?

# The Exponential Distribution (pp. 1054–1055)

In many queuing systems the **inter-arrival time**  $A$  is exponentially distributed. That is,  $P(A \leq t) = \int_0^t \lambda e^{-\lambda t} dt$ .

The probability density function of the **exponential distribution** is  $a(t) = \lambda e^{-\lambda t}$ . It is easily shown that  $E(A) = \frac{1}{\lambda}$  and  $\text{var } A = \frac{1}{\lambda^2}$ .



No-memory property of the exponential distribution (p. 1054)

$P(A > t + h \mid A \geq t) = P(A > h)$  for all nonnegative  $t$  and  $h$ .

# The Poisson Distribution (pp. 1055–1056)

## Theorem 1 (p. 1055)

Inter-arrival times are exponential with parameter  $\lambda$  if and only if the number of arrivals to occur in an interval of length  $t$  follows a Poisson distribution with parameter  $\lambda t$ .

A discrete random variable  $N$  has a **Poisson distribution** with parameter  $\lambda$  if, for  $n = 0, 1, 2, \dots$ ,  $P(N = n) = \frac{e^{-\lambda} \lambda^n}{n!}$  in which case  $E(N) = \lambda$  and  $\text{var } A = \lambda$ .

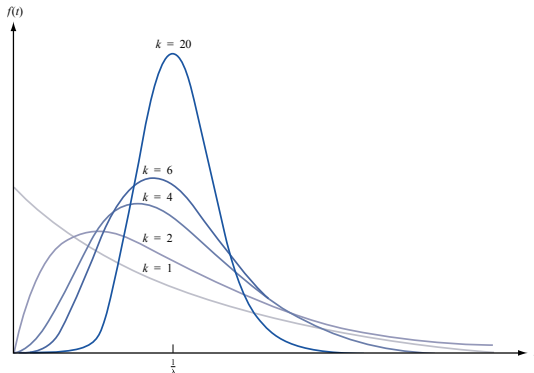
## Example 1, pp. 1057–1058 (Beer Orders)

The number of glasses of beer ordered per hour at a Pub follows a Poisson distribution, with an average of 30 beers per hour being ordered.

- 1 Find the probability that exactly 60 beers are ordered between 10 P.M. and 12 midnight.
- 2 Find the mean and standard deviation of the number of beers ordered between 9 P.M. and 1 A.M.
- 3 Find the probability that the time between two consecutive orders is between 1 and 3 minutes.

# The Erlang Distribution (pp. 1058–1059)

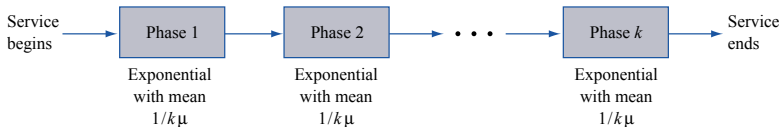
The probability density function of the **Erlang distribution** with rate parameter  $R$  and shape parameter  $k$  is  $f(t) = \frac{R(Rt)^{k-1}e^{-Rt}}{(k-1)!}$ . It is easily shown that  $E(A) = \frac{k}{R}$  and  $\text{var } A = \frac{k}{R^2}$ .



Note that we retain the exponential distribution as special case when  $R = \lambda$  and  $k = 1$ .

# The Erlang Distribution (pp. 1058–1059)

A **service time** that is **Erlang distributed** with rate parameter  $k\mu$  and shape parameter  $k$  is applicable to a queuing system in which service consists of passage through  $k$  phases of service, in which the time to complete each phase has the no-memory property and a mean of  $\frac{1}{k\mu}$ .



Note that the service time may in some cases sometimes be *deterministic*.

1/2/3/4/5/6

Characteristic **1** specifies the nature of the arrival process:

- $M$  = Inter-arrival times are *independent and identically distributed* (iid) random variables having an exponential distribution.
- $D$  = Inter-arrival times are iid and deterministic.
- $E_k$  = Inter-arrival times are iid Erlang distributions with shape parameter  $k$ .
- $GI$  = Inter-arrival times are iid and governed by some general distribution.

1/2/3/4/5/6

Characteristic 2 specifies the nature of the service times:

- $M$  = Service times are iid and exponentially distributed.
- $D$  = Service times are iid and deterministic.
- $E_k$  = Service times are iid Erlang distributions with shape parameter  $k$ .
- $G$  = Service times are iid and follow some general distribution.

Characteristic 3 specifies the number of parallel servers.

1/2/3/4/5/6

Characteristic 4 describes the queue discipline:

- FCFS = First come, first served.
- LCFS = Last come, first served.
- SIRO = Service in random order.
- GD = General queue discipline.

Characteristic 5 specifies the maximum allowable number of customers in the system (including customers who are waiting and customers who are in service).

Characteristic 6 is the size of the population from which customers are drawn.

# Kendall-Lee Notation Example (p. 1061)

The Kendall-Lee notation

$$M/E_2/8/FCFS/10/\infty$$

might represent a health clinic with eight doctors, exponential inter-arrival times, two-phase Erlang service times, a first come, first served queue discipline, and a total capacity of 10 patients.

## Additional Example

The number of students arriving per minute at a classroom follows a Poisson distribution, with an average of 5 students arriving per minute.

- 1 Find the probability that fewer than 3 students arrive within the next 30 seconds.
- 2 Find the probability that more than 2 students arrive within the next 30 seconds.
- 3 Find the mean and standard deviation of the number of students arriving between 11:57 and 11:59.
- 4 Find the probability that the time between two consecutive students arriving is between 10 and 20 seconds.